7.1 Report Definition

To fulfill the primary business goals of enabling real-time management visibility (Milestone 1) and evaluating operational performance improvements over the legacy spreadsheet system (Milestone 2), GreenGrid Utilities makes use of Salesforce's built-in reporting engine.

Reports are organized through specific report types that span both standard and custom object boundaries, ensuring accurate data aggregation without redundancy.

7.1.1 Core Operational Report Specifications

a. Report 1: Average Handling Time (AHT) and Case Triage Performance

- **Report Type:** Cases (Standard Report Type)
- **Format:** Summary Report (Grouped by Owner and Priority)
- **Core Metrics:** Average Age (Hours/Minutes), Total Case Count, and Case Status Breakdown.
- **Business Justification:** Directly monitors BR1 (targeting a 25% reduction in AHT). It reveals how efficiently agents log, process, and transfer incoming utility outage calls within the Salesforce Service Console, compared to the 5-minute delay caused by the legacy spreadsheet system (B1).

b. Report 2: Field Visit Resolution Efficiency

- **Report Type:** Cases with Field Visits (Custom Report Type built on a Case parent and Field_Visit__c child relationship)
- **Format:** Matrix Report (Rows grouped by Field_Visit__c.Owner / Field Technician; Columns grouped by Field_Visit__c.Repair_Status__c)

- **Core Metrics:** Record Count (Total Dispatches), Field Visit Success Rate (Ratio of Resolved vs. Failed), and Average Turnaround Time.
- **Business Justification:** Evaluates field operational performance (BR3 — targeting a 15% improvement in first-visit fix rate) and tracks the transition away from informal, untracked messaging platforms (B2).

c. Report 3: High-Priority Grid Outage Trends

- **Report Type:** Cases with Utility Assets (Custom Report Type built on a Case parent and Utility_Asset__c child relationship)
- **Format:** Summary Report (Grouped by Utility_Asset__c.Asset_Status__c and Asset Installation Address/Region)
- **Core Metrics:** Total High-Priority Cases, Outage Duration Tracking, Asset Failure Frequency.
- **Business Justification:** Gives Executive Leadership instant visibility into which critical grid hardware components are experiencing repeated service disruptions.

7.2 Dashboard Design

Dashboards serve as the central command interface for GreenGrid Utilities' leadership tier (Apex Tier in the Role Hierarchy). They transform rows of raw report data into real-time, easily scannable visual performance components.

7.2.1 Executive Operations Dashboard Blueprint

Component 1 (Top Left) – Open Crisis Metric

A large single-number metric component displaying the active count of High-Priority Cases where Status = New or Working. It refreshes instantly whenever Flow 1 (Dispatch Engine) runs.

Component 2 (Top Center) – Service Console Speed Gauge

A visual gauge tracking Average Case Age. The green zone represents optimal handling speed below 2 minutes, directly demonstrating the resolution of spreadsheet bottleneck B1.

Component 3 (Top Right) – First-Visit Fix Ratio Donut

A donut chart displaying the breakdown of Field_Visit__c.Repair_Status__c values (Assigned, In Progress, Resolved, Failed). This measures the elimination of paper log losses (B3).

Component 4 (Bottom Main) – Grid Outage Distribution Bar Chart

A vertical bar chart tracking Case counts grouped by Asset Installation Address. This supports executive planning for physical infrastructure improvements.

7.3 Data Visualization Strategy

To maintain the performance and configuration standards outlined in NFR3 (Dashboards must load within a 5-second window), GreenGrid Utilities enforces strict data visibility rules and access controls for dashboards.

7.3.1 Role Hierarchy Integration & Dashboard View Context

- Dashboards are set to run under a designated administrative user context ("View Dashboard As: Specified User" assigned to an Executive Manager).
- This setup allows the system to pre-cache complex multi-object roll-ups across the entire organization. It avoids the performance delays caused by real-time role-based sharing recalculations during live presentations, in accordance with NFR3 load-time requirements.

7.3.2 Foldering and Security Governance

- Reports and dashboards are stored within a restricted folder environment named **Executive Operational Analytics**.
- **Access Control:** The folder is granted **Viewer** access to the Customer Support Team and Field Technicians role groups to reduce workspace clutter, and **Manage/Editor** access to Executive Management and IT/System Administrators.

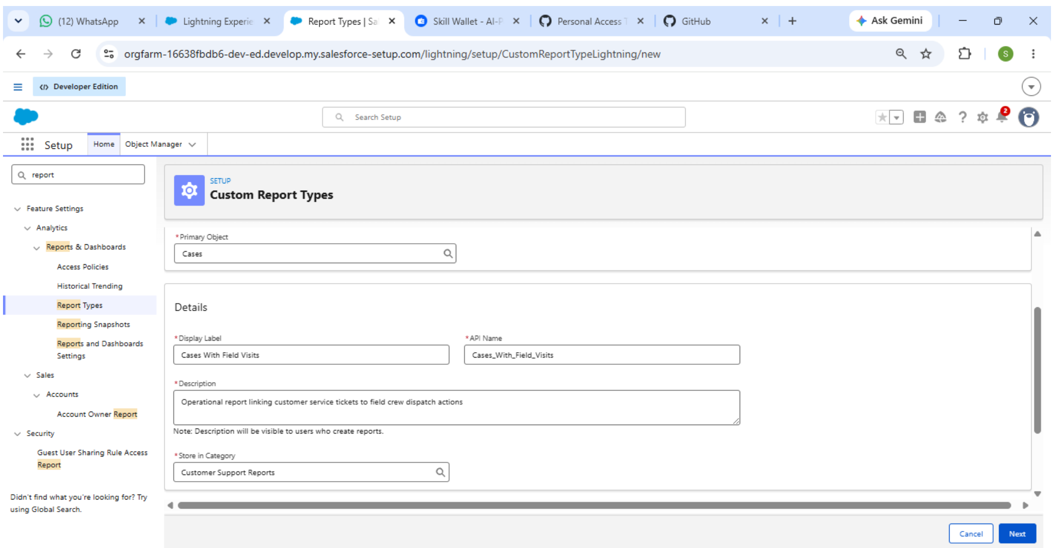
7.4 Salesforce Implementation and Configuration Steps

Follow these step-by-step instructions to define custom report structures, build analytical components, assemble the executive dashboard layout, and secure the operational folders.

Step 1: Create Custom Report Types for Multi-Object Relationships

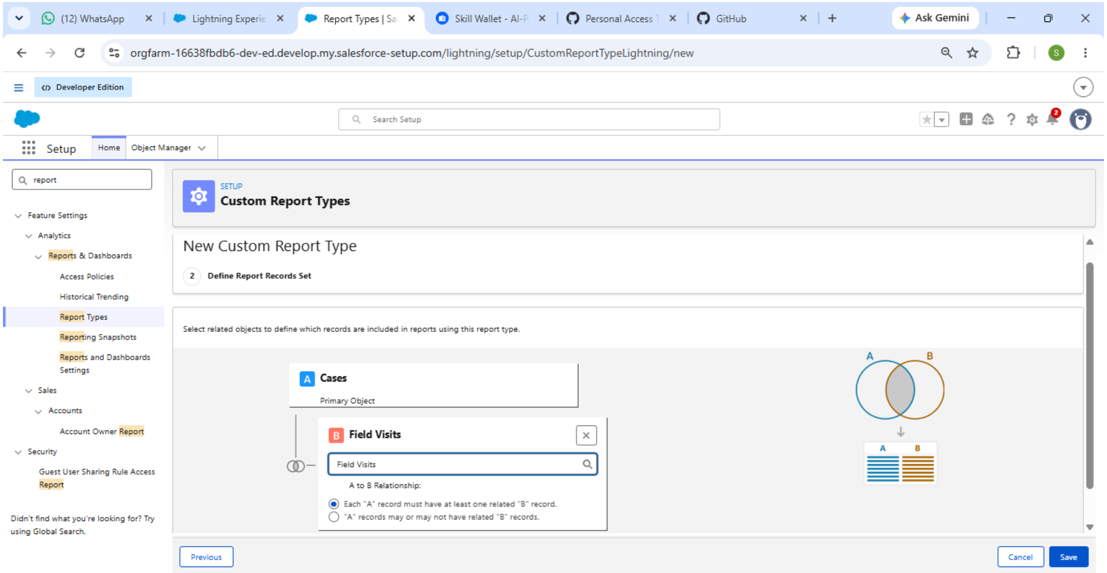
Before building reports, custom frameworks must be deployed to connect custom entities (Utility_Asset__c and Field_Visit__c) with standard objects.

- Navigate to **Setup** by clicking the gear icon in the top-right corner.
- In the Quick Find box on the left, type **Report Types** and select **Custom Report Types**.
- If an introductory screen appears, click **Continue**.
- Click **New Custom Report Type**.
- Fill in the Report Type settings as follows:
    - **Primary Object:** Select Cases.
    - **Report Type Label:** Type Cases with Field Visits.
    - **Report Type Name:** Auto-populates as Cases_with_Field_Visits.
    - **Description:** Type *Operational report linking customer service tickets to field crew dispatch actions.*
    - **Store in Category:** Select Customer Support Cases.
    - **Deployment Status:** Select Deployed.

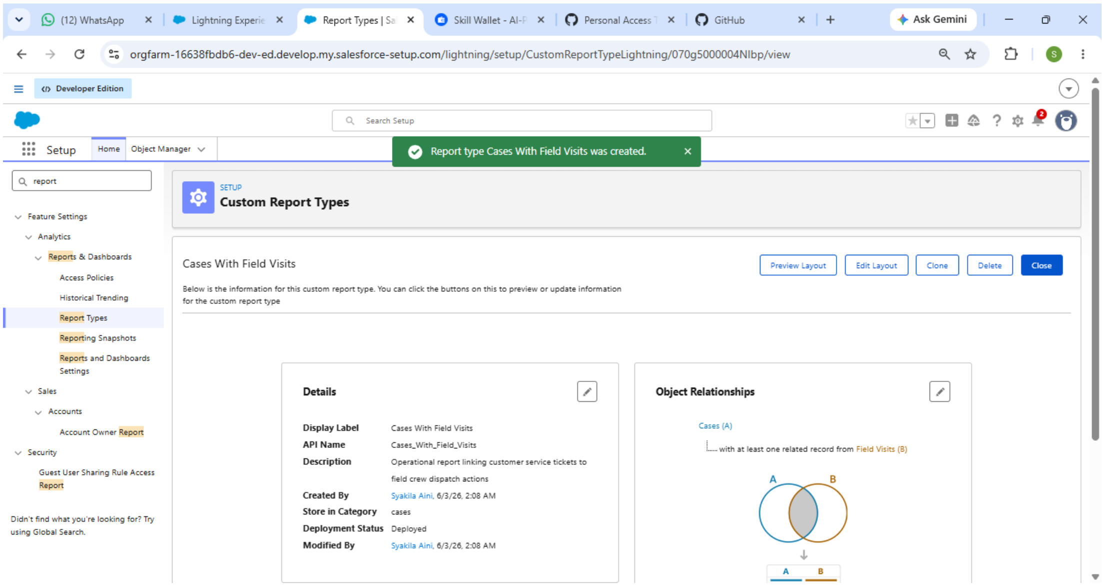


- Click **Next**.
- On the Define Object Relationships canvas, click the empty box below A (Cases) to add a child object.

- Select **Field Visits (Field_Visit__c)** from the dropdown. Leave the default setting: *Each "A" record must have at least one related "B" record.*



- Click **Save**.

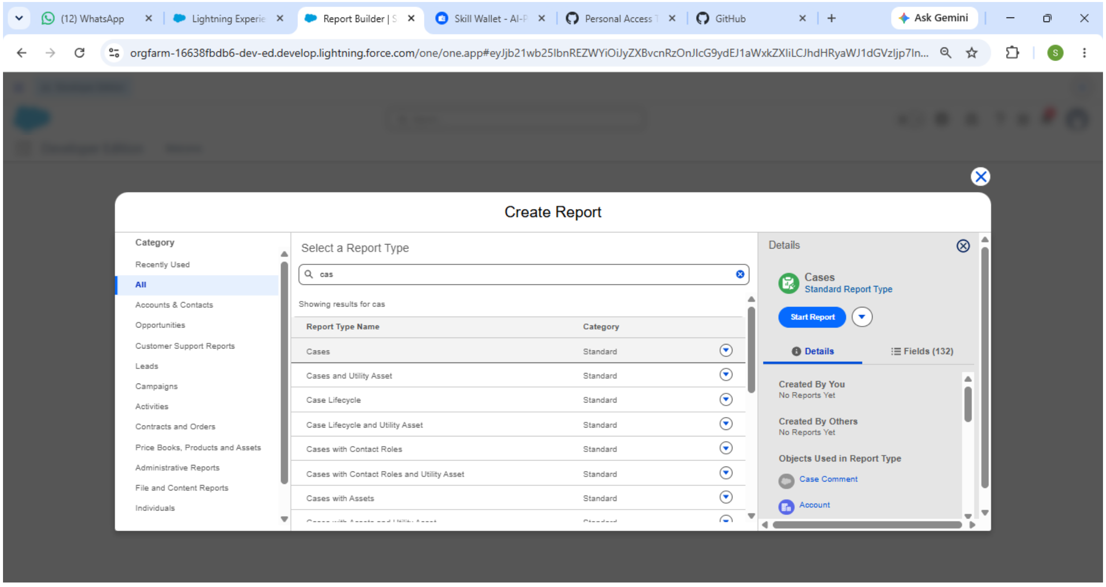


- *(Optional Repeat for Assets)* Click **New Custom Report Type** again. Create a type with Cases as Primary, link Utility Assets (Utility_Asset__c) as object B, label it **Cases with Utility Assets**, set it to Deployed, and click **Save**.

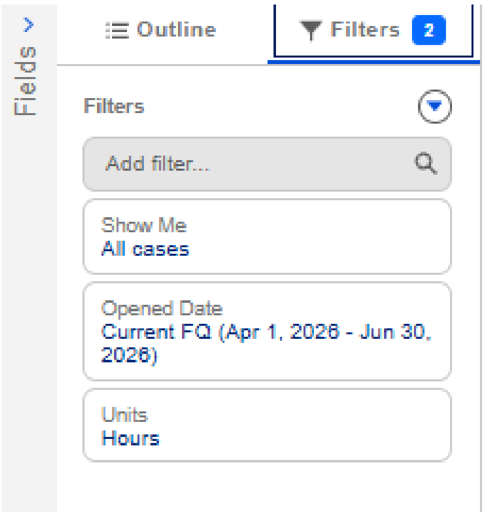
Step 2: Build Report 1 — Average Handling Time (AHT) Analytics

- Click the **App Launcher** icon and search for **Reports**.
- Click **Reports**, then click **New Report**.

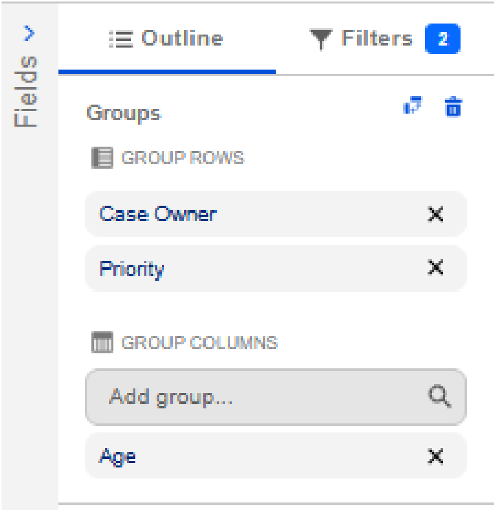
- Search for and select the standard **Cases** report type.
- Click **Start Report**.



- Enable **Update Preview Automatically**.
- Under **Filters:**
    - Change **Show Me** from My Cases to **All Cases**.
    - Set **Opened Date** to Current FQ (or All Time for testing).

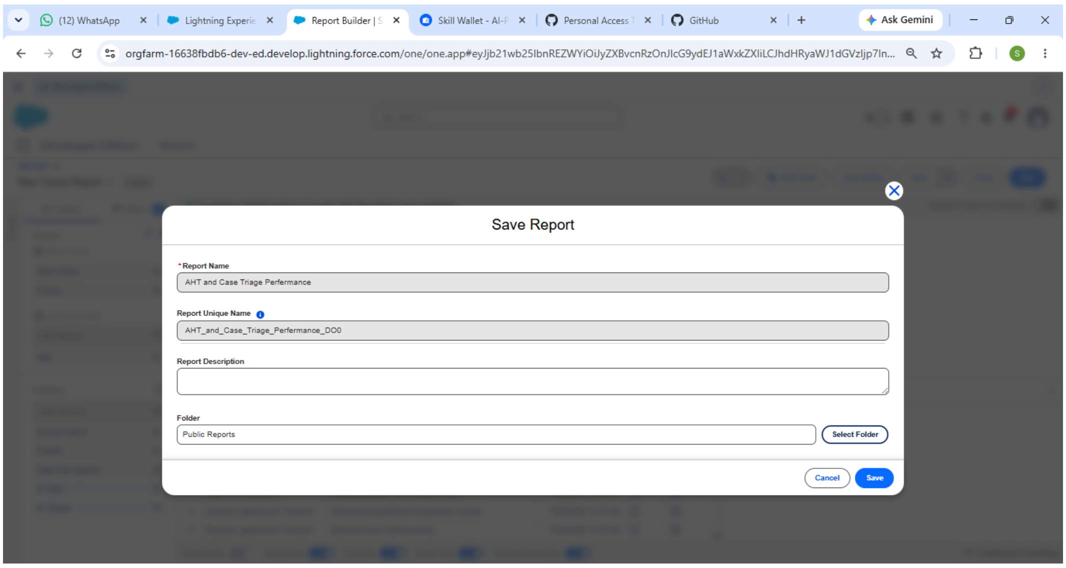


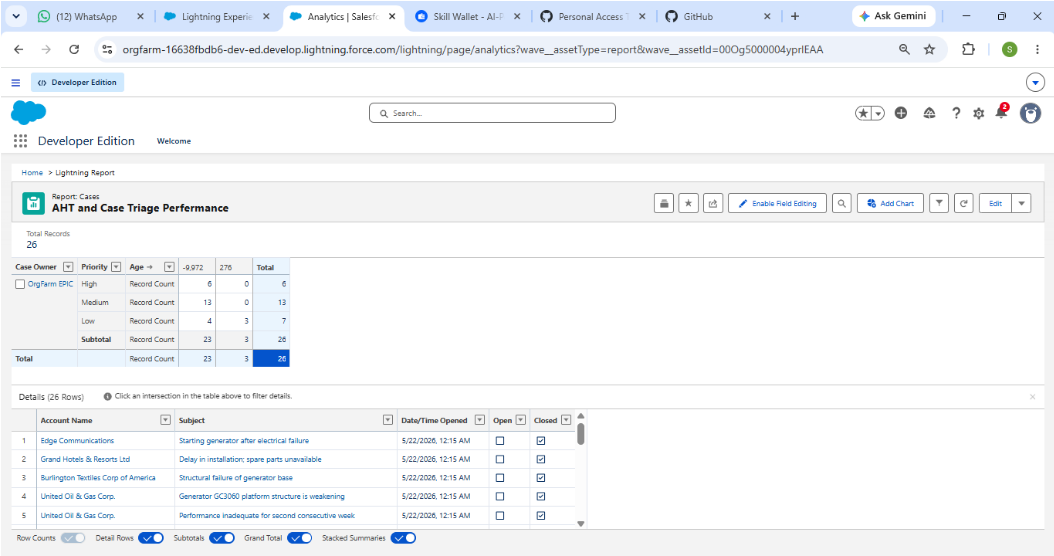
- Under **Outline:**
    - Group Rows by **Case Owner**.
    - Add a second grouping by **Priority**.



- Add the **Age** column and remove unnecessary fields.
- Summarize the Age field using **Average**.

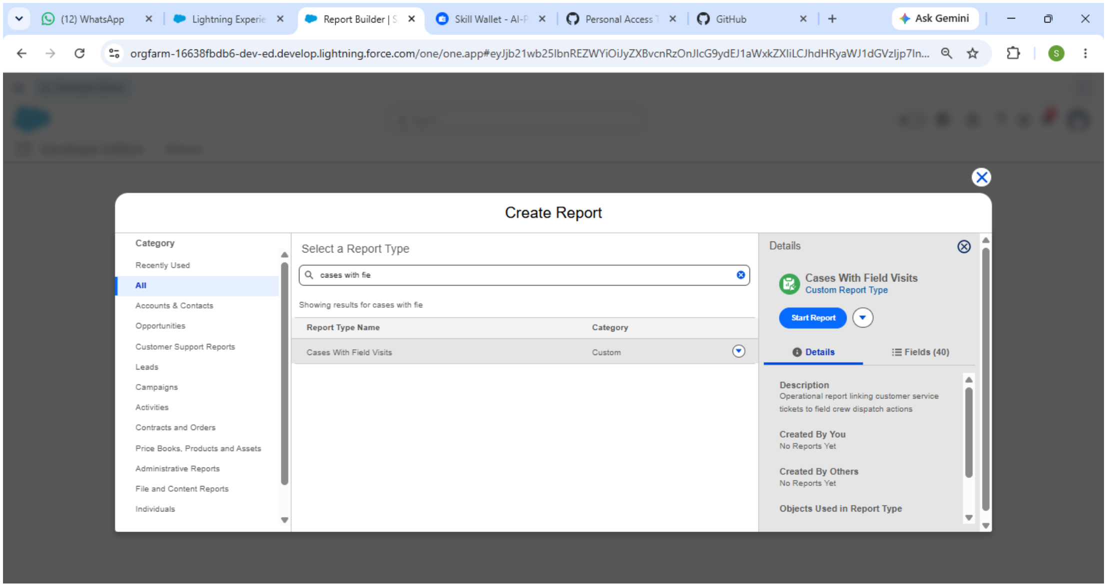
- Click **Save & Run**.
- Save as:
  - **Report Name:** AHT and Case Triage Performance
  - **Unique Name:** AHT_and_Case_Triage_Performance



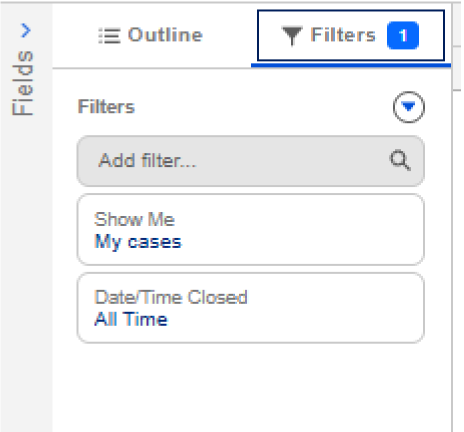


Step 3: Build Report 2 — Field Visit Resolution Efficiency

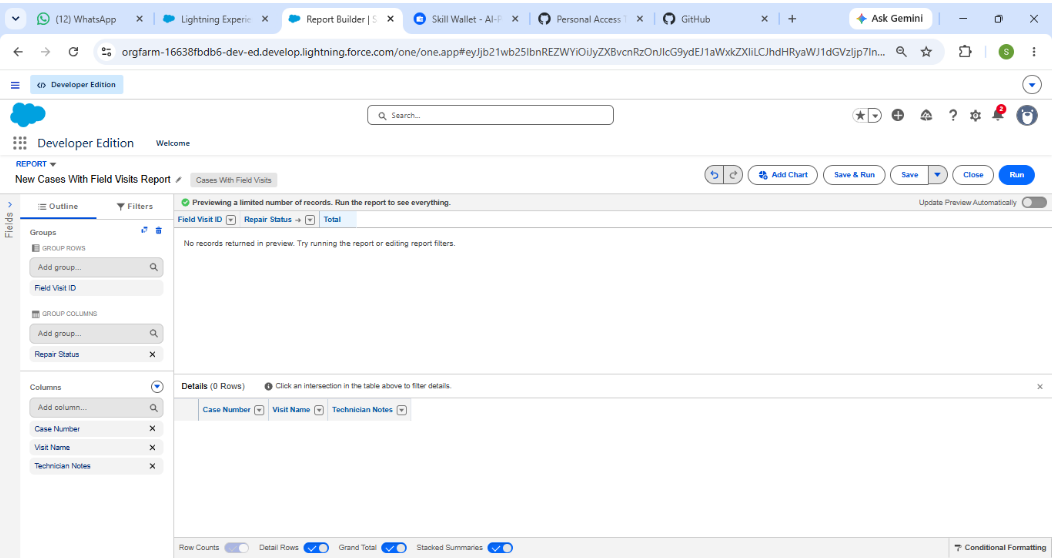
- Click **New Report**.
- Select **Cases with Field Visits**.



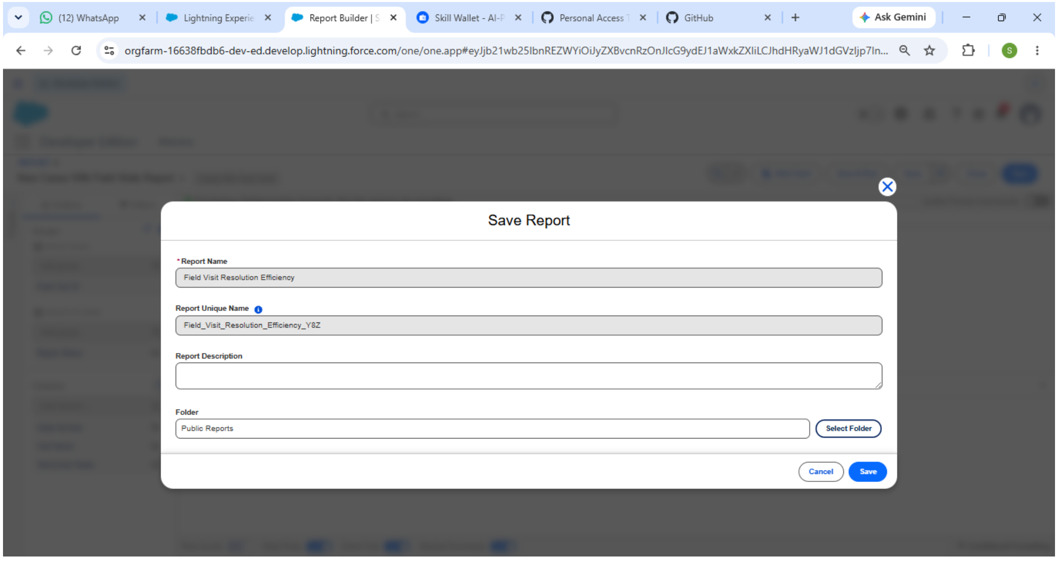
- Click **Start Report**.
- Under **Filters:**
    - **Show Me:** All Cases
    - **Opened Date:** All Time



- Under **Outline:**
  - Group Rows by **Field Visit Owner**
  - Group Columns by **Repair Status**
  - Add: **Visit Number** and **Technician Notes**

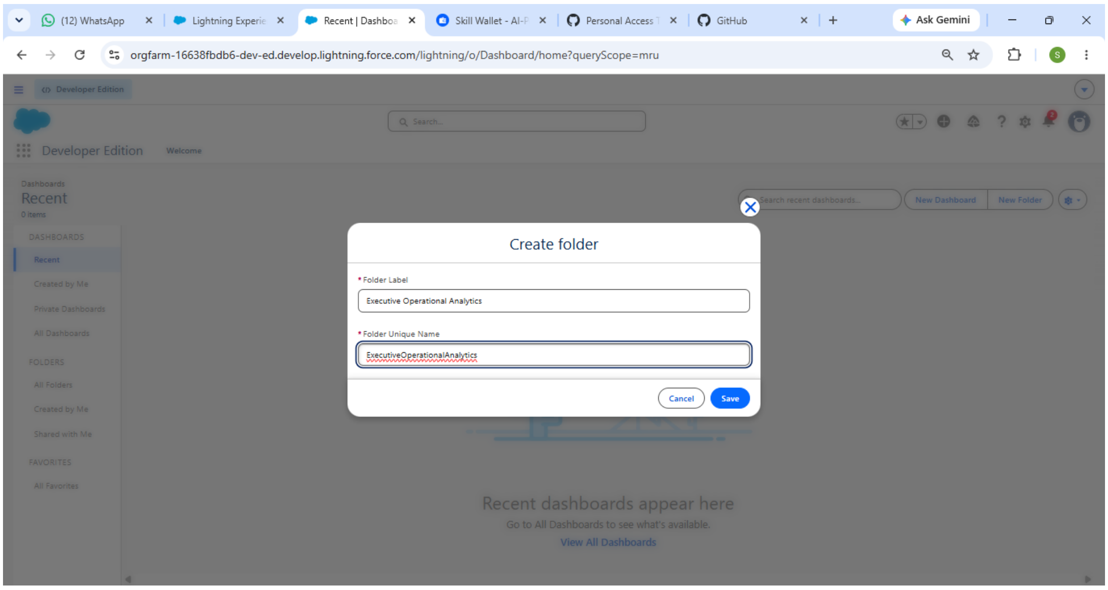


- Click **Save & Run**.
- Save as:
  - **Report Name:** Field Visit Resolution Efficiency
  - **Unique Name:** Field_Visit_Resolution_Efficiency

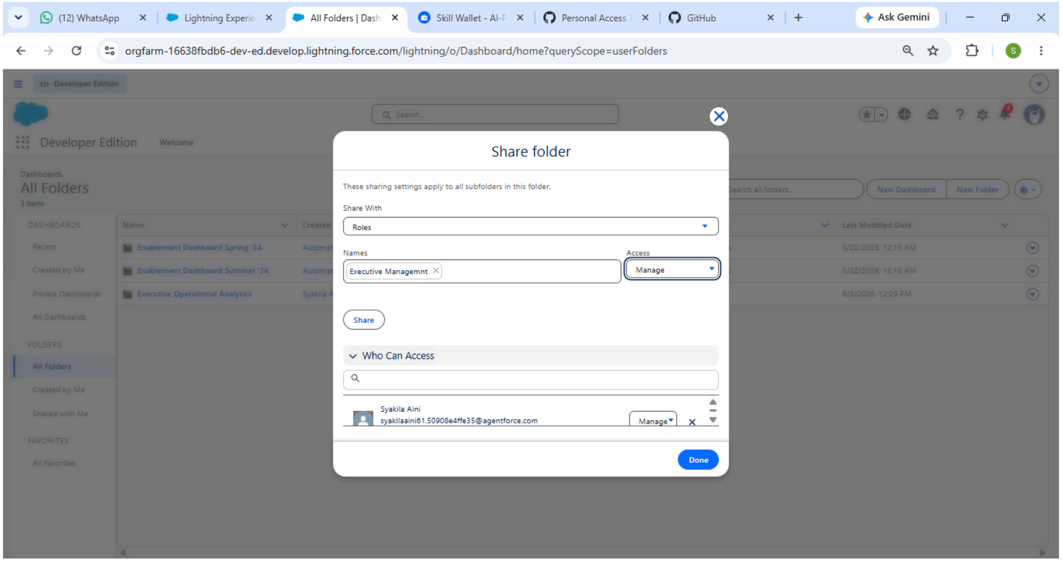


Step 4: Provision Secured Analytics Folders

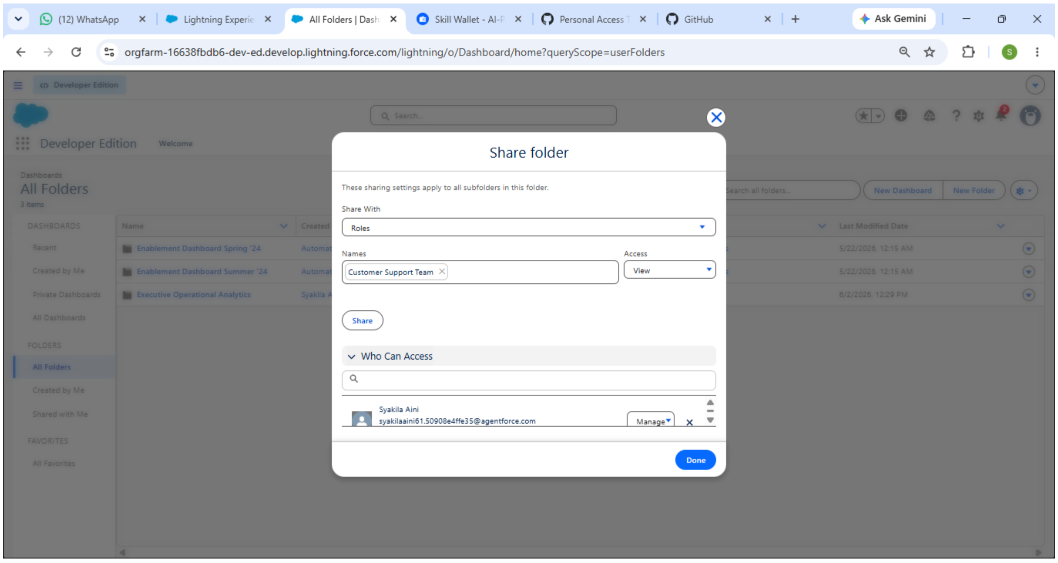
- Open the **Dashboards** tab.
- Click **New Dashboard Folder**.
- Name the folder: **Executive Operational Analytics**.



- Click **Save**.
- Open **Share** settings for the folder.
- Configure:
  - **Executive Management** → Manage



- **Customer Support Team** → Viewer

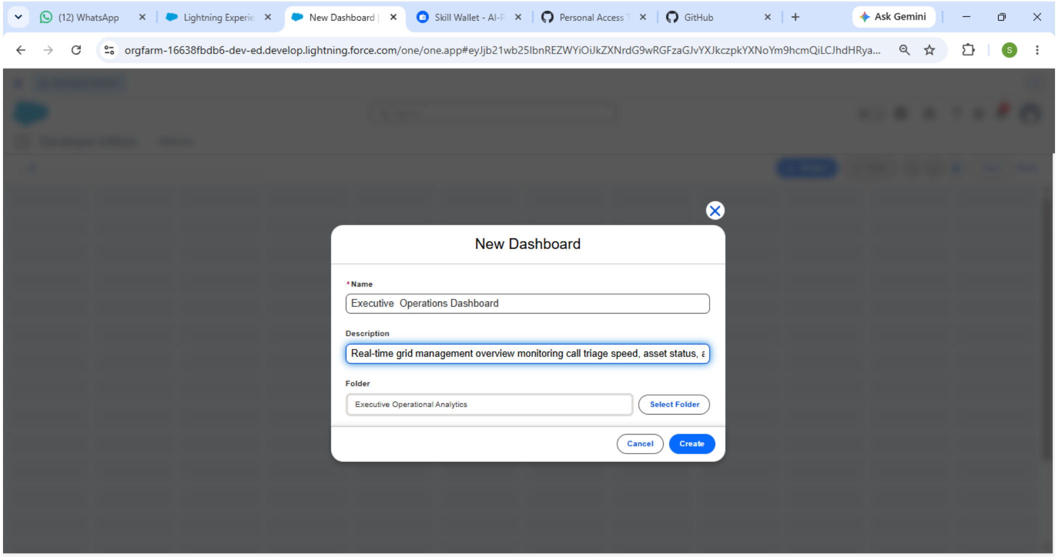


- Click **Done**.

Step 5: Assemble the Executive Operations Dashboard

- Open **Dashboards**.
- Click **New Dashboard**.
- Configure:
  - **Name:** Executive Operations Dashboard
  - **Description:** Real-time grid management overview monitoring call triage speed, asset status, and mobile technician field repair rates.

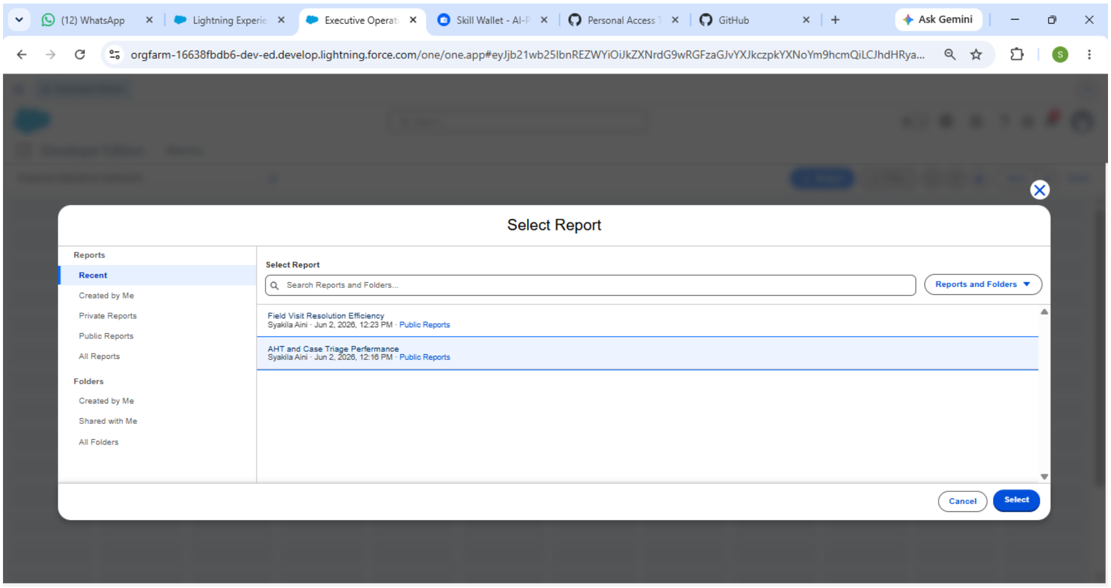
- **Folder:** Executive Operational Analytics



- Click **Create**.

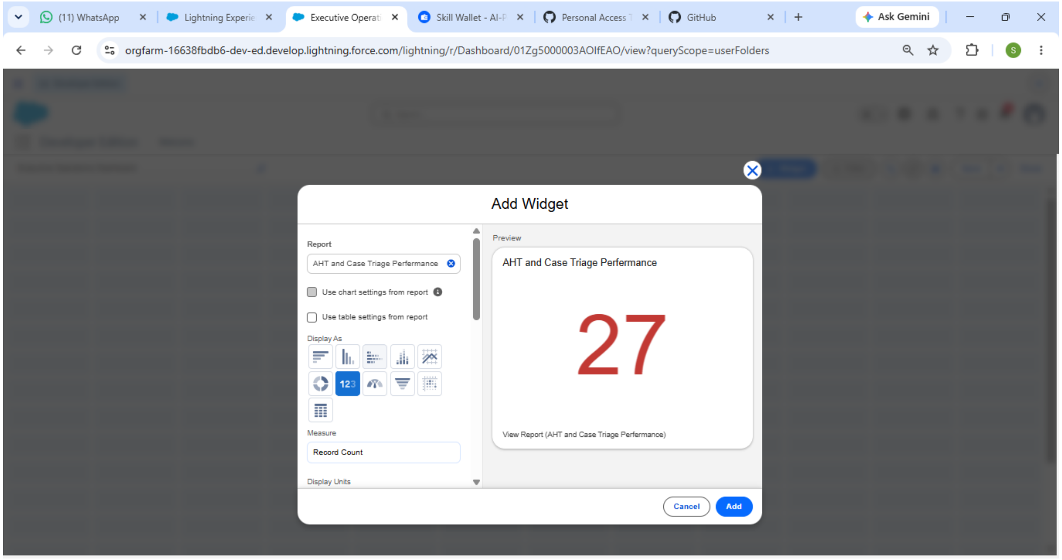
Step 6: Add Component 1 — Live Open Crisis Counter

- Click + **Component**.
- Select report: **AHT and Case Triage Performance**.



- Configure:
  - **Component Type:** Metric
  - **Measure:** Record Count

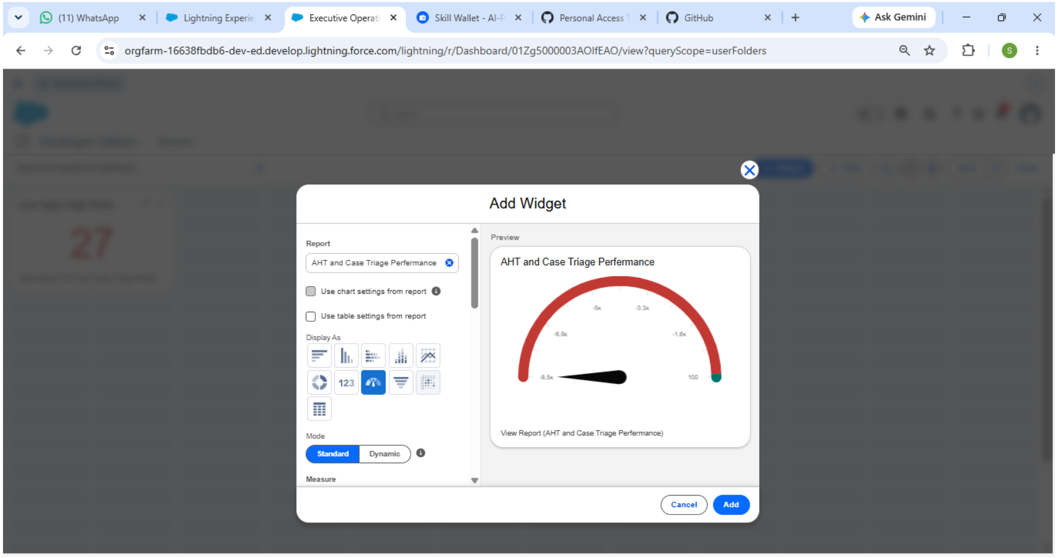
- **Title:** Live Open High-Priority Cases



- Click **Add** and position it in the upper-left section.

Step 7: Add Component 2 — Service Console Speed Gauge

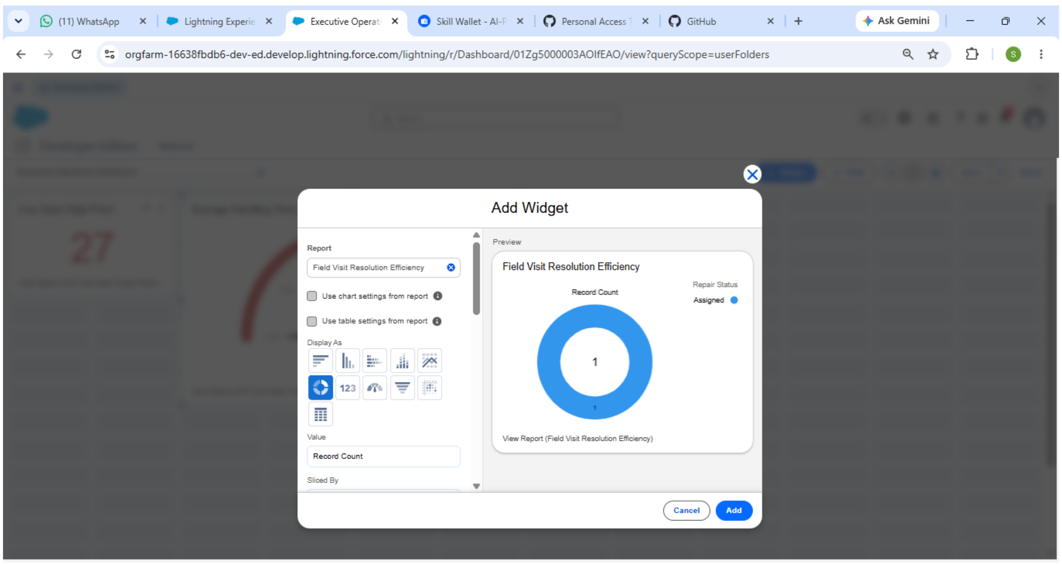
- Click + **Component**.
- Select report: **AHT and Case Triage Performance**.
- Configure:
  - **Component Type:** Gauge
  - **Measure:** Average Age
  - **Breakpoint 1:** 2
  - **Breakpoint 2:** 5
  - **Title:** Average Handling Time (AHT) Gauge



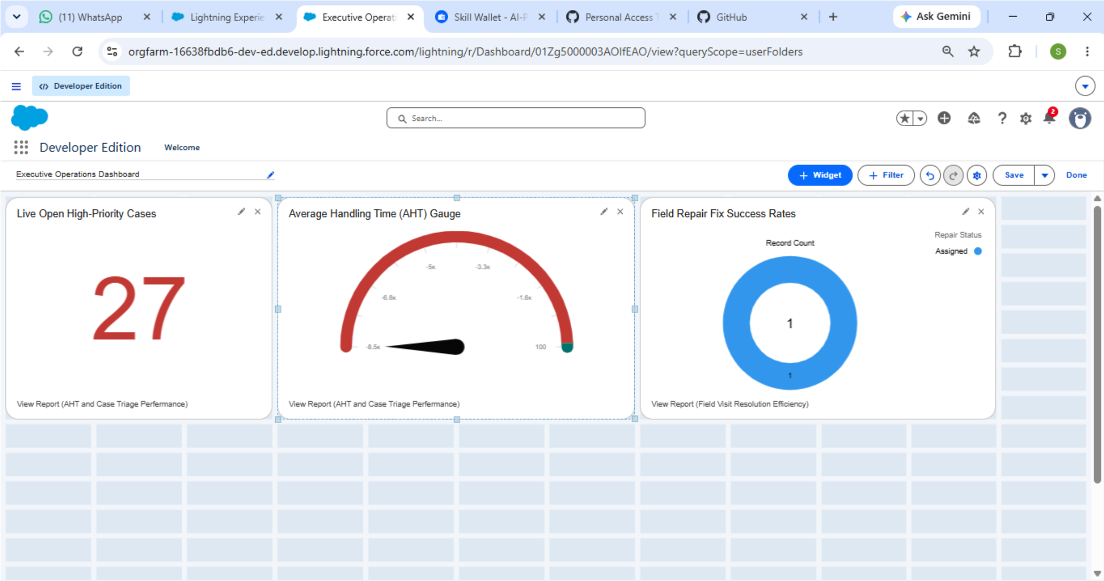
- Click **Add** and position it in the top center.

Step 8: Add Component 3 — First-Visit Fix Ratio Donut

- Click + **Component**.
- Select: **Field Visit Resolution Efficiency**.
- Configure:
  - **Component Type:** Donut Chart
  - **Metric:** Record Count
  - **Slice By:** Repair Status
  - **Title:** Field Repair Fix Success Rates

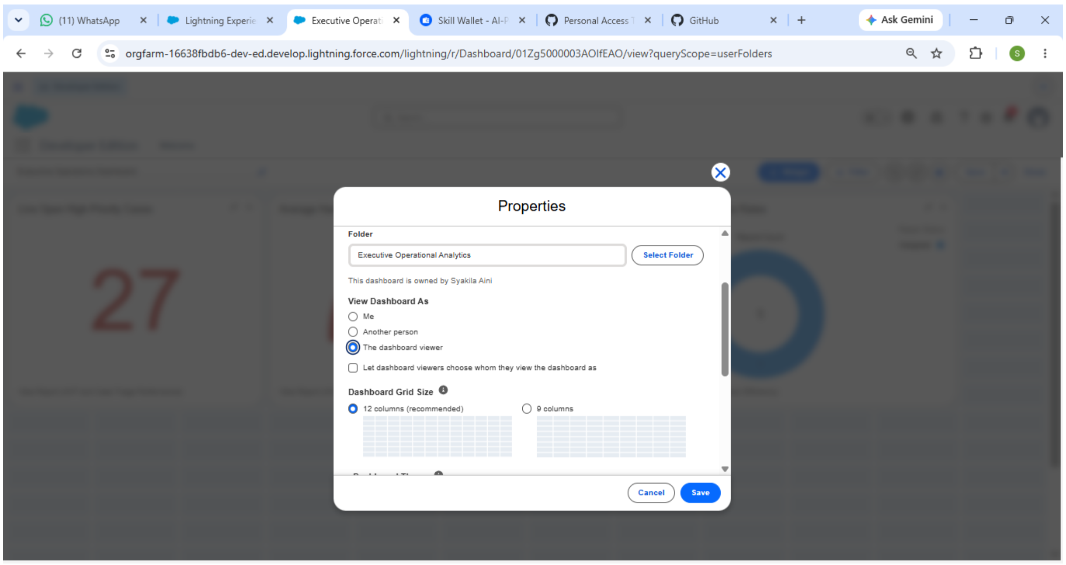


- Click **Add** and position it in the upper-right section.



Step 9: Configure Dashboard Run Context and Finalize Activation

- Open **Dashboard Settings**.
- Under **View Dashboard As:**
  - Choose **The Dashboard Viewer**, or
  - Select an **Executive Management** user.



- Click **Save**.
- Save the dashboard.

- Click **Done** to launch the active dashboard.

Conclusion

With the completion of Milestone 7: Reporting and Analytics Design, GreenGrid Utilities has put in place a data-driven layer on top of its CRM architecture.

By mapping custom object relationships (Utility_Asset__c and Field_Visit__c) into structured reports and real-time dashboard metrics, the organization can now track service speed improvements, confirm reductions in handling times, and assess field technician efficiency. Secured analytics folders ensure that sensitive operational performance data is delivered reliably and efficiently to executive leadership.